



Miracle Millets: Climate-resilient crops for nutritional security and sustainable agriculture

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Introduction

Small millets are cereal crops grown for their tiny grains on short, slender grassy plants, including finger, kodo, foxtail, proso, little and barnyard millets. Finger millet is considered a small millet but not a minor millet. These crops are staple foods for millions living in the arid and semi-arid tropics across Asia, Africa and parts of Europe. Small millets are resilient, providing reliable harvests in low-rainfall areas and during potential famine conditions. Nutritionally, they are superior to rice and wheat, offering affordable proteins, minerals and vitamins to vulnerable populations. They are also resistant to storage pests, giving them a long shelf life. With their high nutritional value and untapped yield potential, small millets are gaining recognition as “miracle grains” or “superfoods” and are poised to become important future crops, especially in rainfed and climate-challenged regions.

These crops are adapted to wide range of temperatures, moisture regimes and low input conditions supplying food and feed to millions of dryland farmers, particularly in the developing world. Due to the climate change, the average global temperature on earth has increased by about 0.8°C since 1880. In addition to the aberrant monsoon patterns, rise in CO₂ levels paired with higher temperatures also posing a major threat to the world food production. Millets possess C₄ photosynthesis system wherein with less photorespiration efficiently utilizes the scarce moisture situation of arid and semi-arid regions. Most cereals like rice or maize cannot support temperatures over 35°C during grain filling stages, whereas pearl millet can fill its grain even up to 42°C. In spite of all these extraordinary qualities and capacities of millets as climate resilient crops, area under its cultivation has been shrinking over the last five decades since green revolution due to relentless promotion of other crops such as rice, wheat and maize for intensive farming under irrigated or rainfed conditions. Contribution of millets in total food grain production of India reduced from 22.17 per cent to 6.94 per cent over the last six decades from 1950-51 to 2011-12.

In India, millets are cultivated in an area of 15.48 m ha producing 17.2 m t with a yield of 1111 kg ha⁻¹. Diminishing land and other natural resources provides us with an option of crop intensification to feed the ever-growing population. As millets are fast maturing, they can easily fit into intensive cropping systems which contribute to annual food grain production.

Global climate change

Climate refers to the long-term patterns of weather, including temperature, humidity, precipitation, wind and atmospheric pressure. Climate change is the long-term alteration of these patterns, either in the average climate or its variability, typically persisting for decades or longer. It adversely affects sectors like agriculture, hydropower, forestry and biodiversity. In agriculture, climate change can influence crop productivity, growth rates, photosynthesis, transpiration and soil moisture availability.

Global mean temperatures have risen by approximately 0.74°C over the past century, primarily due to greenhouse gas emissions from fossil fuel use and land-use changes. Projections indicate a further rise of 1.8-6.4°C by 2100, with greater warming during the rabi season. Changes in precipitation patterns are expected, with declines during December-February, while snow cover is likely to shrink and extreme events such as heatwaves and heavy rainfall may become more frequent. In India, crop production could decline by 10-40% by 2080-2100, with regions like western Rajasthan, southern Gujarat, Madhya Pradesh, Maharashtra, northern Karnataka, northern Andhra Pradesh and southern Bihar being particularly vulnerable to extreme climatic events (Mall *et al.*, 2006).

Smart foods

Smart foods of the 21st century are ones which are good for you, the planet and the farmer to cope up changing climate scenario as presented in the table.

Good for humans	Good for the planet	Good for the farmer
▪ Multiple health benefits ▪ High in dietary fiber, antioxidants, zinc, folic acid, protein, calcium <i>etc.</i> ▪ Three times higher calcium ▪ Diet diversity	▪ Millets are fast growing putting less stress on the environment ▪ Food solution in changing climate ▪ Low water foot print ▪ Can grow with minimal fertilizers and pesticides	▪ Multiple uses – food, fodder, biofuel and brewing ▪ Withstand drought conditions ▪ Heat tolerant ▪ Greater yield gain potential

Special characteristics of minor millets

- Fast maturing - fit them into intensive cropping systems
- Drought adaptation
- Grains can be stored well for long periods - ensures food supply
- Many ways of processing - increasing market demand
- Dual purpose - food and fodder
- Lower cost of production
- Improved agro-biodiversity
- Mutually beneficial for intercropping
- Nutritionally superior to other major cereals

Drought adaptation in millets

Drought escape: Drought escape refers to the condition in which plants reach maturity before the drought occurs. Traits associated with drought escape are rapid growth, early flowering, high leaf nitrogen level and high photosynthetic capacity. The study in West Africa indicated that pearl millet matches its phenology to the mean distribution of the rainfall where precipitation is limited and erratic. In this case, the development of the main panicle coincided with an increasing period of rain, thus reducing the risks associated with drought events occurring prior to or at the beginning of flowering.

Drought avoidance: Drought avoidance refers to the ability of the plant to maintain a favourable water balance under moisture stress in order to avoid water deficit in the plant tissue.

Two types of drought avoidance mechanisms have been identified:

- (i) those that reduce water loss through transpiration (e.g. low stomata conductance and reduced leaf).
- (ii) those that maintain water uptake during drought period (e.g. high root-to-shoot ratio).

Drought tolerance: Drought tolerance refers to the ability of the plant to produce some yield by withstanding low water potential. Traits associated with drought tolerance are increased osmo-protectants (or compatible solutes such as amino acids) and osmotic adjustment (i.e. reducing osmotic potential through accumulation of organic and inorganic substances).

Mechanisms of drought response in diverse millet types

Parameter	Millet type	Response to drought
Agronomy-related traits		
Seed number and biomass	Pearl millet	Unaffected under drought
Seed yield	Pearl millet	High for drought-tolerant genotypes
Flowering time	Pearl millet	Adjust phenology to rainfall pattern
Morphology-related traits		
Shoot length	Little millet	Decreased under drought
Root length	Little millet	Increased under drought
Leaf tensile strength	Tef	Increased in drought-tolerant plants
Physiology-related traits		
Water extraction	Pearl millet	Less extraction before flowering; more extraction after flowering
Biochemical-related traits		
Anti-oxidants	Little millet	Accumulated under drought
ROS scavenging enzymes	Little millet, tef	Accumulated under drought
Free proline	Little millet, tef	Increased concentration
GB (glycine betaine)	Little millet	Accumulated under drought
Superoxide	Little millet	Accumulated under drought
AP (ascorbate peroxidase)	Tef, little millet	Increased specific activity

Crop intensification

With rising food demand, climate change and limited soil and water resources, farmers face the challenge of producing more with less. The principles of the System of Rice Intensification (SRI), originally developed to improve irrigated rice productivity, have been adapted to other crops under the broader concept of System of Crop Intensification (SCI). SCI aims to optimize the use of available resources like soil, water, seeds, nutrients, sunlight and air to increase crop productivity and resilience. In Haveri district, Karnataka, farmers have developed innovative practices for growing finger millet that closely resemble SRI management techniques, demonstrating the potential of SCI for diverse crops.

Conclusion

Minor and small millets are climate-resilient, nutritionally rich and highly adaptable crops that can play a key role in ensuring food security under changing climatic conditions. Their fast growth, drought tolerance, low input requirements and long shelf life make them ideal for rainfed and resource-constrained regions. Integrating millets into intensive cropping systems through crop intensification strategies can enhance productivity, improve agro-biodiversity and support sustainable agriculture. Given their health benefits, environmental sustainability and suitability for farmers, millets represent a “smart food” solution for addressing the challenges of climate change, population growth and malnutrition in India and beyond.

References

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CATALOGUES